



TAMPINES MERIDIAN JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION

H2 BIOLOGY

Paper 3 Long Structured and Free-response Questions

9744/03

16 September 2022

2 hours

SUGGESTED ANSWERS

Section A

Answer all questions in this section.

QUESTION 1

All life on earth is categorized into three domains – Eukarya, Bacteria and Archaea.

- Eukarya: organisms with cells that contain a nucleus as well as membrane-bound organelles
- Bacteria: unicellular prokaryotic cells that do not contain a nucleus and membrane-bound organelles
- Archaea: a unique class of unicellular prokaryotic cells that present many distinctive features from the domain Bacteria

Archaea were first discovered in 1977 in high-temperature environments, such as hydrothermal vents and terrestrial hot springs. They were later also found to be residing in other extreme conditions, such as highly acidic and anaerobic environments.

(a) One feature unique to Archaea is their cell membranes. Fig. 1.1 shows the molecular structure of the membranes of Eukarya and Bacteria and the membranes of Archaea.

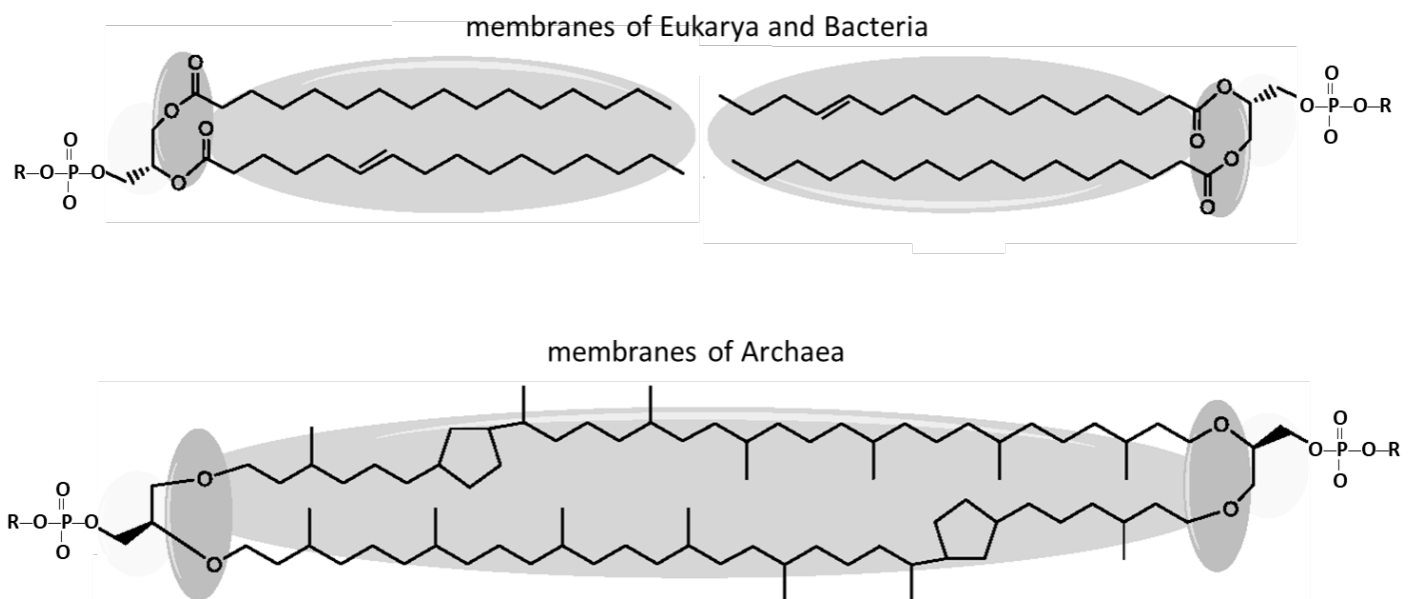


Fig. 1.1

- (i) Describe the structural differences between the membranes of Eukarya and Bacteria and the membranes of Archaea. [3]

Eukarya & Bacteria	Archaea
1. a <u>bilayer</u>	<u>monolayer</u>
2. hydrocarbon chain <u>unbranched</u>	hydrocarbon chain <u>branched</u>
3. Absence of <u>ring/cyclic/pentagonal</u> hydrocarbon	Presence of <u>ring/cyclic/pentagonal</u> hydrocarbon
4. Presence of <u>C=C double bonds</u> OR <u>unsaturated</u> hydrocarbons	Only <u>C–C single bonds</u> OR <u>saturated</u> hydrocarbons
5. <u>Less carbons</u> (due to no branching)	<u>More carbons</u> (due to branching)
6. <u>Shorter</u> hydrocarbon chain	<u>Longer</u> hydrocarbon chain
7. <u>Ester</u> bond between glycerol and hydrocarbon chains	<u>Ether</u> bonds between glycerol and hydrocarbon chains

[Any 3]

- (ii) Suggest how **one** of the structural differences you have described in (a)(i) enables Archaea to survive in high-temperature environments. [2]

- Monolayer/cyclic: allows the membrane to be less fluid / more rigid / **AW**
OR Branched/more carbons: increases hydrophobic interactions
OR Lacks C=C double bonds, hence no kinks, hence less fluid
- At high temperatures, allow membranes to remain intact / not rupture / **AW**

- (b) Some archaeal cells synthesize ATP via photophosphorylation, using ATP synthase and a protein called bacteriorhodopsin. Permanently associated with bacteriorhodopsin is a light-harvesting prosthetic group called retinal, which changes conformation upon light absorption. On top of its light-harvesting property, bacteriorhodopsin is also a proton pump.

Fig. 1.2 shows the structure of bacteriorhodopsin and ATP synthase embedded in the archaeal cell surface membrane.

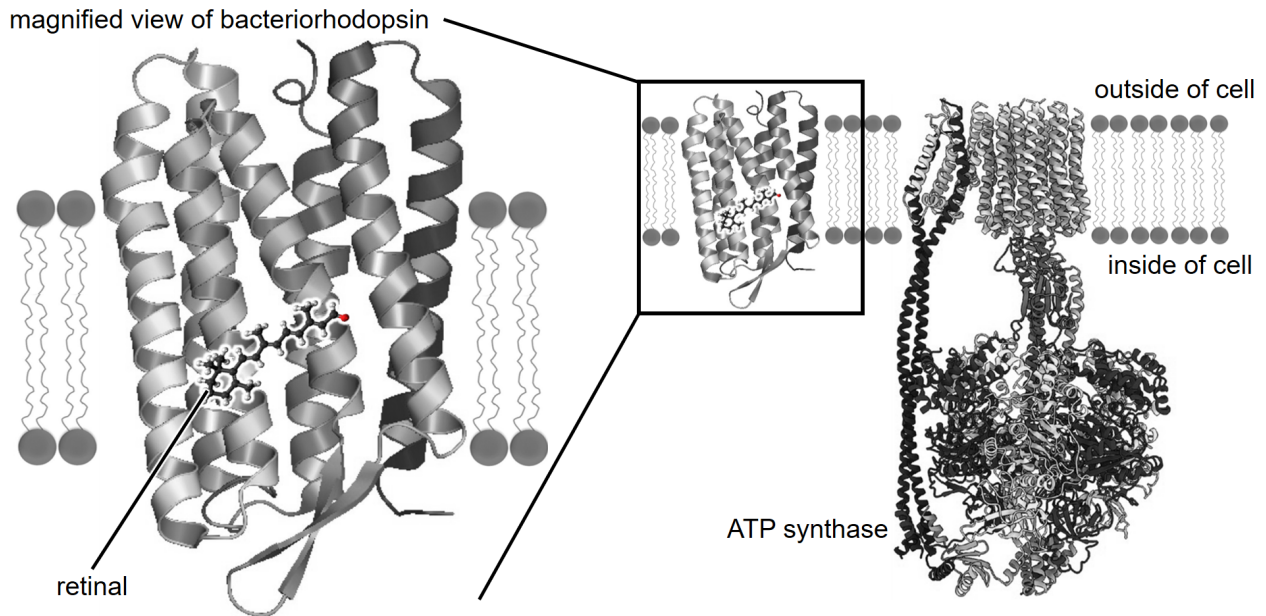


Fig. 1.2

- (i) Outline the structural arrangement of bacteriorhodopsin in the archaeal membrane. [2]

1. Seven α -helices...
2. ...that span the membrane / a transmembrane protein
3. β -pleated sheet faces the inside of cell
4. Retinal molecule within the membrane

[Any 2]

- (ii) With reference to Fig. 1.2, suggest how photophosphorylation occurs in the Archaea. [4]

1. Retinal absorbs light energy and changes conformation
2. Induces conformational change in bacteriorhodopsin, activating it.
3. Bacteriorhodopsin pumps protons from inside of archaea to outside, establishes proton gradient (across archaea membrane)
4. Facilitated diffusion of protons through ATP synthase (embedded in archaeal surface membrane) into the cell
5. Proton-motive force used to drive ATP synthesis from ADP and inorganic phosphate

[Any 4]

- (c) Before their discovery in 1977, archaeal cells had been wrongly classified as bacterial cells, receiving the name Archaeobacteria, for many years.

Fig. 1.3 shows two possible evolutionary relationships among the three domains of life.

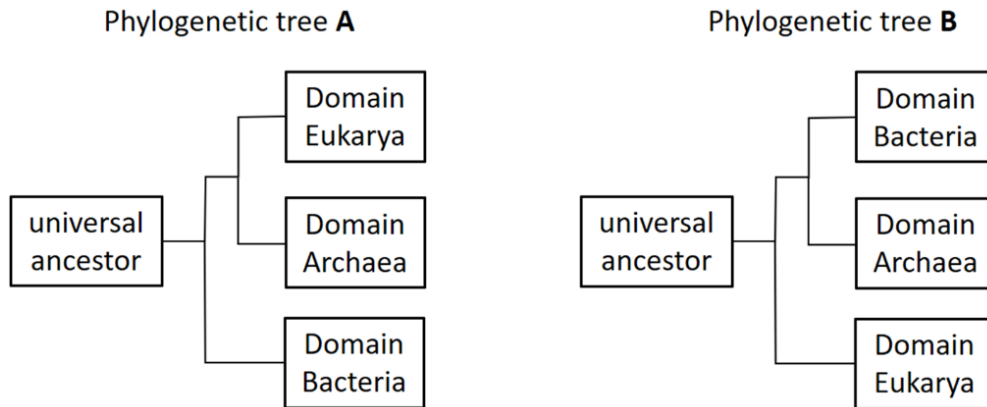


Fig. 1.3

Fig. 1.4 shows the general structure of an archaeal cell.

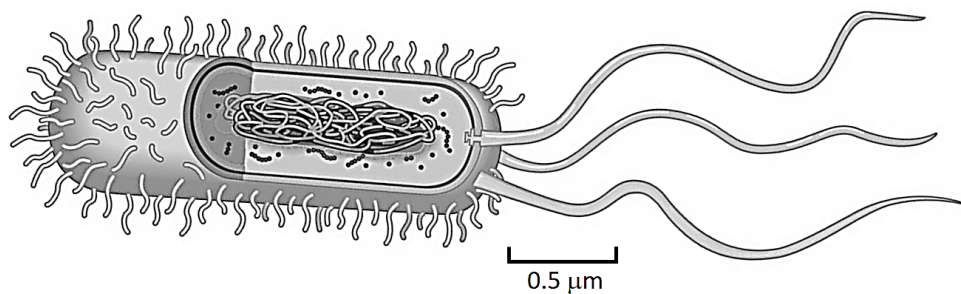


Fig. 1.4

- (i) With reference to Fig. 1.3 and Fig. 1.4, explain why Phylogenetic Tree **A** could appear to be unexpected. [3]

1. **Idea that** Archaea is morphologically very similar to Bacteria
2. **Ref to** any one structural feature shared with Bacteria, e.g. size, no nucleus, multiple flagella, cell wall, pili, **AVP**
3. Hence Bacteria and Archaea are expected to be more closely related (Tree **B**)
OR
Tree **A** shows that Eukarya is more closely related to Archaea, hence unexpected.

- (ii) Although the domains Bacteria, Archaea, and Eukarya were founded based on genetic criteria, biochemical properties have also provided strong evidence that Archaea form an independent prokaryotic group and that they share features with both Bacteria and Eukarya. Major examples of these features are shown in Table 1.1.

Table 1.1

Features	Bacteria	Archaea	Eukarya
organization	unicellular	unicellular	unicellular and multicellular
nuclear envelope	absent	absent	present
DNA form	single, circular	single, circular	multiple, linear
histones associated with DNA	absent	present	present
nature of promoter	Pribnow box	TATA box	TATA box
presence of introns	no	in some genes	yes
types of RNA polymerase	Bacterial in nature	Eukarya-like	Eukaryal in nature
transcription factors	absent	present	present
operon	present	present	absent
initiator amino acid during translation	formyl-methionine	methionine	methionine
ribosomes	70S	Eukarya-like	80S
peptidoglycan in cell wall	present	absent	absent
response to the antibiotics streptomycin and chloramphenicol	inhibited	not inhibited	not inhibited

Explain which Phylogenetic Tree (**A** or **B**) is supported by the data in Table 1.1. [3]

1. Tree A
2. Archaea shares more similarities (9 out of 13) with Eukarya than with Bacteria
3. **Idea that** Archaea and Eukarya shares more molecular homologies, which is the basis for phylogeny
4. **One example:** promoter sequences, genes for transcription factors / histones / RNA polymerases / rRNA / ribosomal proteins

[Point 1 + Any 2]

- (iii) One of the structural similarities shared between Archaea and Eukarya is the use of histone proteins in DNA packaging. Fig. 1.5 shows how DNA is packaged in eukaryotic and archaeal cells.

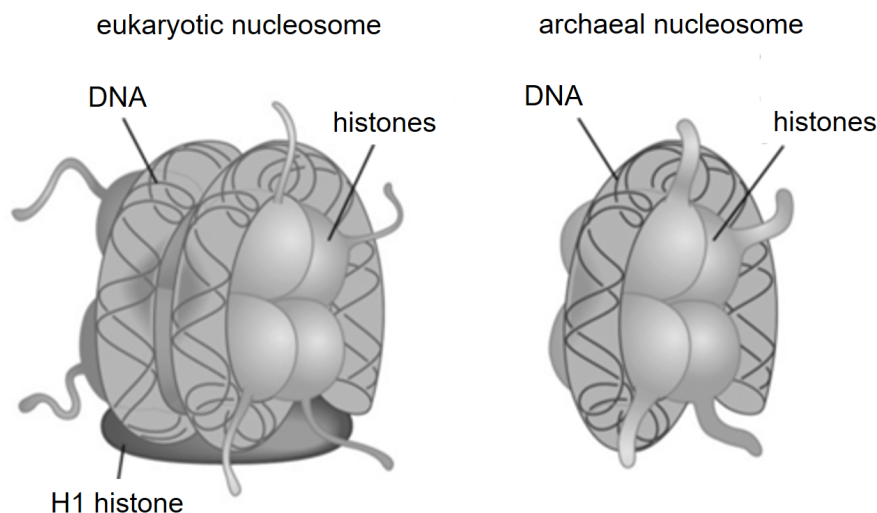


Fig. 1.5

State the differences in DNA packaging between eukaryotic and archaeal cells. [2]

eukaryotic	archaeal
1. presence of <u>H1 histone</u>	absence of H1 histone
2. <u>eight histones</u> in <u>each nucleosome</u>	<u>four histones</u> in <u>each nucleosome</u>
3. <u>two turns</u> of DNA per nucleosome	<u>one turn</u> of DNA per nucleosome

[Any 2]

- (d) In eukaryotic cells, proteins destined for secretion are first synthesized by ribosomes bound to the rough endoplasmic reticulum (ER) membrane. The synthesized polypeptides then enter the ER lumen, where they are packaged into transport vesicles that undergo exocytosis. Exocytosis is not possible in archaeal and bacterial cells due to the lack of internal membranes.

Using the above information, suggest how archaeal and bacterial cells secrete proteins. [3]

- Ribosomes bind directly (to channel proteins) on the cell surface membrane
- Idea that** Emerging polypeptide then traverse through the cell surface membrane...
- ...via protein channel embedded on the cell surface membrane
- Folds into globular shape only when outside the cell

[Any 3]

- (e) Although some Archaea inhabit the human body and possess some characteristics of human pathogens, surprisingly, no pathogenic archaeal species has been identified to date.

Characteristics possessed by human pathogens that are shared by Archaea include:

- They are highly diverse and are present in large numbers in the environment that would afford them the opportunity to cause disease.
 - They are recognized by antigen-presenting cells, B cells and phagocytes.
 - Some possess genes that code for toxin proteins needed for pathogenicity.
- (i) Some scientists have partly attributed the lack of pathogenic archaeal cells to the observation that there are far less viruses that infect Archaea than viruses that infect Bacteria.

Suggest why scientists have made this attribution. [2]

1. Viruses serve as a vehicle for gene transfer via transduction
2. Virulent/toxin genes can be transferred from one species of archaea to another
OR
 Viruses can contain virulent/toxin genes that can be directly transferred to bacteria
3. **Idea that** Lack of archaeal viruses hence limits genetic diversity of pathogenic archaea

[Any 2]

- (ii) Using Table 1.1 on page 7, suggest why the emergence of pathogenic archaeal cells could threaten the entire human population. [1]

- Existing antibiotics (e.g. streptomycin and chloramphenicol, **from Table 1.1**) are not effective against archaeal cells

- (f) The recent discovery of giant viruses shatters the textbook definition of viruses as “filterable” infectious agents because these giant viruses do not pass through bacterial filters. This destroys all boundaries between viruses and cellular life forms in terms of size.

Not only are giant viruses larger than numerous bacterial and archaeal cells, but the genomes of Pandoraviruses, the current record holder at approximately 2500 kilobases, are also larger and more diverse in gene content than many bacterial and archaeal genomes. Several scientists hence have proposed the need to classify viruses into a fourth domain of life, but such proposal was met with many disagreements from other scientists.

Fig. 1.6 shows the relationship between virus particle size and genome size from all the different virus families. The size of the circles represents of the relative average viral particle size from each virus family.

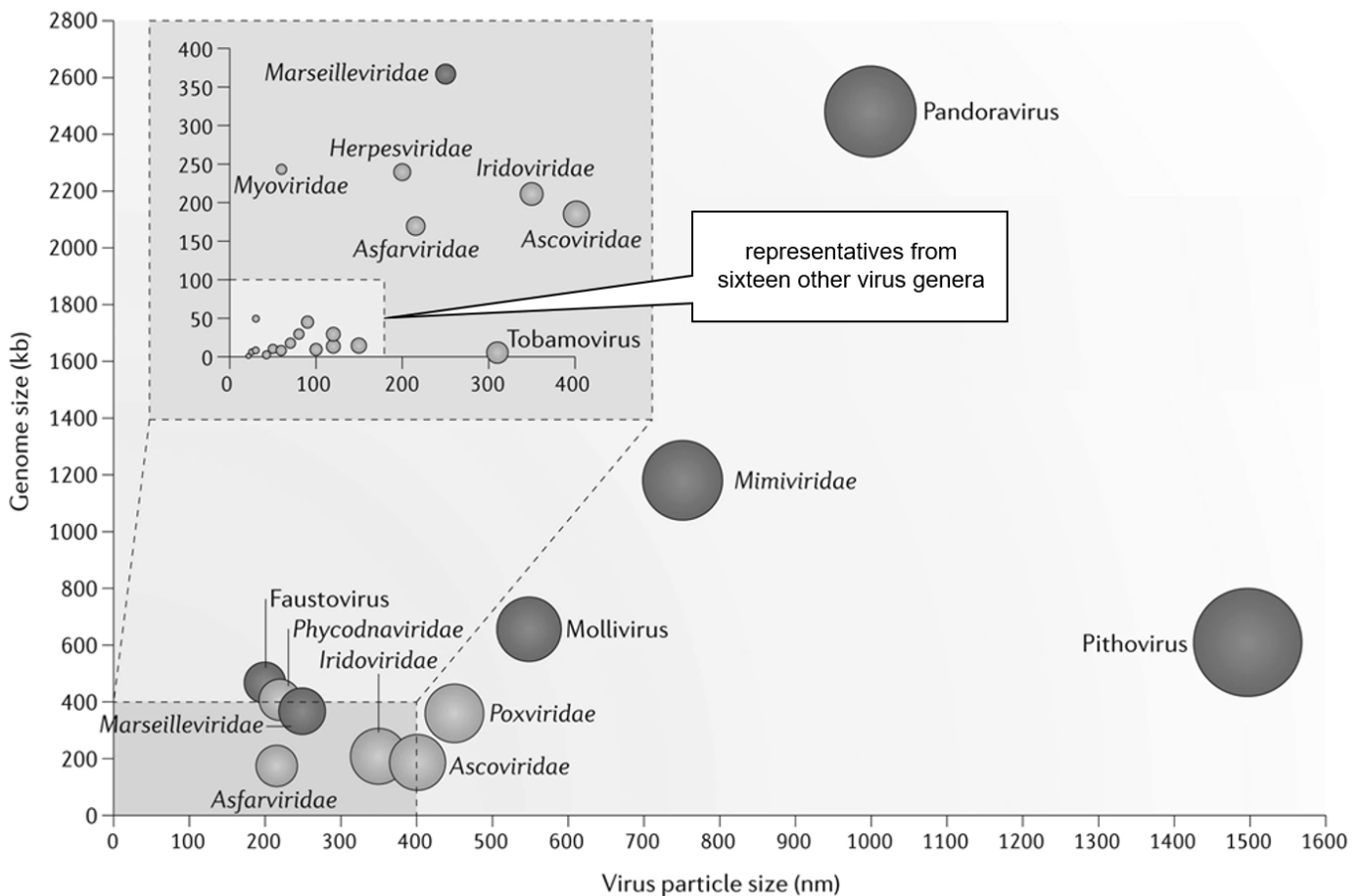


Fig. 1.6

- (i) Comment on the trend data in Fig. 1.6. [3]

1. **[Compulsory]** In general, as particle size increases, genome size increases
2. **Valid data for the increasing trend**
3. Anomaly: Tobamovirus, 310nm but genome size is only 5kb.
4. Anomaly: Pithovirus, 1500nm but genome size is only 600kb.
5. Anomaly: Myoviridae, 50nm but genome size is 245kb.
6. Most virus falls within particle size 400nm and within genome size 400kb
7. **AVP**

[Any 3]

(ii) Suggest why the proposal to classify viruses into a fourth domain of life has been a subject of controversy. [3]

1. **[COMPULSORY]** Viruses are considered non-living, hence does not fit into the domain of *life*
2. Cannot reproduce/replicate independently on their own
3. Need to exploit the metabolic machinery of host cells for reproduction
4. Lacks cellular enzymes / **named example** (e.g. glycolytic enzymes) for metabolic reactions / **named example** (e.g. respiration) to produce ATP
5. Lacks organelles / **named example** (e.g. ribosomes) for cellular processes / **named example** (e.g. translation)
6. Cannot respond to stimuli / Cannot move on their own
7. **AVP**

[Any 3]

[Total: 31]

QUESTION 2

Proteins must fold into defined three-dimensional structures to gain functional activity. In the cellular environment, newly synthesized polypeptides are at great risk of misfolding and aggregation. Cells hence engage proteins called chaperones to assist in protein folding.

These chaperones have two roles:

1. They bind to proteins to promote folding.
2. They direct misfolded polypeptides for degradation in the cytosol.

However, polypeptides that are in the midst of folding may be mistaken by chaperones as misfolded proteins and hence are directed for degradation. Therefore, protein folding needs to be completed quickly to prevent premature degradation.

A recently discovered endoplasmic reticulum (ER) protein complex called S-E complex was found to delay premature degrading of polypeptides that are in the midst of folding. In its absence, approximately 30% of newly synthesized proteins that could otherwise fold correctly are degraded.

Fig. 2.1 illustrates these processes.

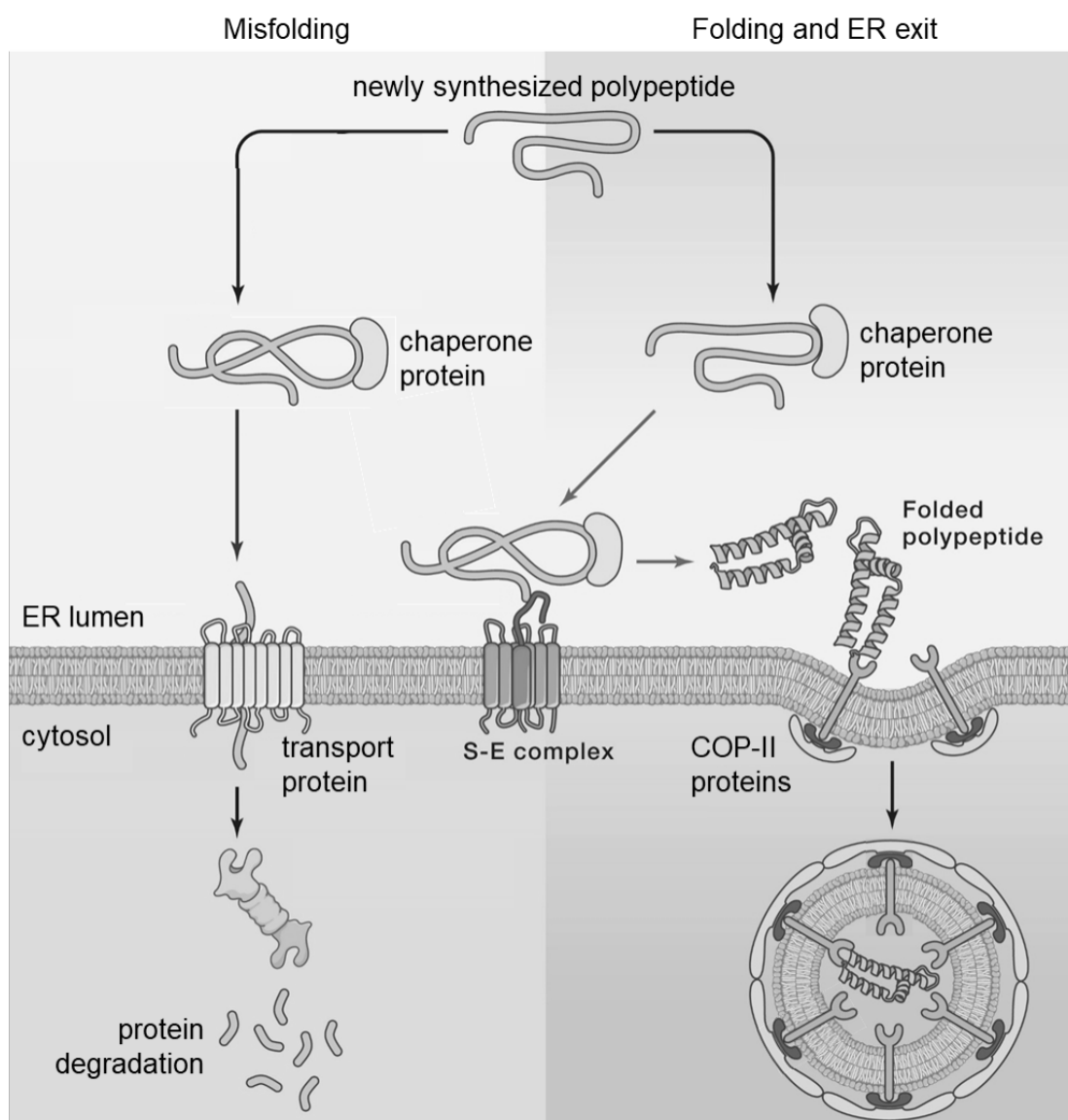


Fig. 2.1

- (a) Protein folding is driven mainly by the primary structure of the protein. In the midst of folding, the R-groups of polypeptides are exposed to their surrounding environment that contains many other different molecules.

Using the information above, suggest why it is important to have chaperone proteins to assist in protein folding. [1]

- **Idea of** shielding the exposed R-group to prevent them from forming bonds with other molecules

- (b) With reference to Fig. 2.1, suggest how the S-E complex allows polypeptides to complete their folding. [2]

1. S-E complex (is a membrane-anchored protein that) binds to chaperone-polypeptide
2. **Idea of** Prevents polypeptide from being prematurely transported out of the ER for degradation
OR
Allows polypeptide to stay longer in the ER to complete folding

- (c) ER vesicle formation is not a random event but is carefully orchestrated.

With reference to Fig. 2.1, describe how ER vesicle formation is triggered. [2]

1. Folded protein binds to the domain of COP-II protein facing the ER lumen
2. Such binding triggers ER membrane to undergo budding / pinching off
3. **Idea that** fusion of the two approaching ends of phospholipid bilayer forms a vesicle

[Any 2]

- (d) With reference to Fig. 2.1 and your knowledge on protein degradation, explain how unfolded polypeptides in the ER are degraded. [3]

1. Polypeptide transported through a transport protein embedded in the RER membrane
2. Attached to ubiquitin proteins (when in cytosol)
3. Targeted to the proteasome
4. For hydrolysis into individual amino acids / shorter peptides

[Any 3]

(e) Misfolded proteins in the ER tend to spontaneously associate with one another to form an aggregate, causing cellular toxicity. It is hence important that any misfolded protein is immediately degraded and not remain in the ER for too long.

(i) Explain why misfolded proteins in the ER tend to spontaneously associate with one another to form an aggregate. [2]

1. Misfolded proteins have their hydrophobic amino acid R groups/side chains exposed
2. **Idea of** The need to shield themselves away from the aqueous ER lumen
3. Hence, hydrophobic R groups from different misfolded protein associate with one another via hydrophobic interaction

[Any 2]

(ii) Suggest why accumulation of protein aggregates in the ER is toxic to the cell. [1]

1. **Idea that** Such aggregate may interact with enzymes / membrane proteins to interfere with their function
2. **Idea that** constrains the ER lumen space available to metabolic enzymes / other proteins
3. **AVP**

[Any 1]

- (f) Protein aggregates are usually due to genetic diseases. One classic example is Huntington's disease, a neurodegenerative disorder with a wide variation in the age of onset. A mutation in the *HTT* gene causes Huntington's disease. The *HTT* gene codes for a protein called huntingtin.

The *HTT* mutation that causes Huntington's disease involves a DNA segment in the *HTT* gene known as a CAG trinucleotide repeat. This segment is made up of a series of CAG that appears multiple times in a row.

An increase in the number of CAG repeats leads to the production of an abnormally long version of the huntingtin protein. The elongated protein is cut into smaller, toxic fragments that aggregate together and accumulate in neurons, disrupting the normal functions of these cells.

Fig. 2.2 shows how the number of CAG repeats varies in 1200 people with different ages of onset. Each data point represents an individual person.

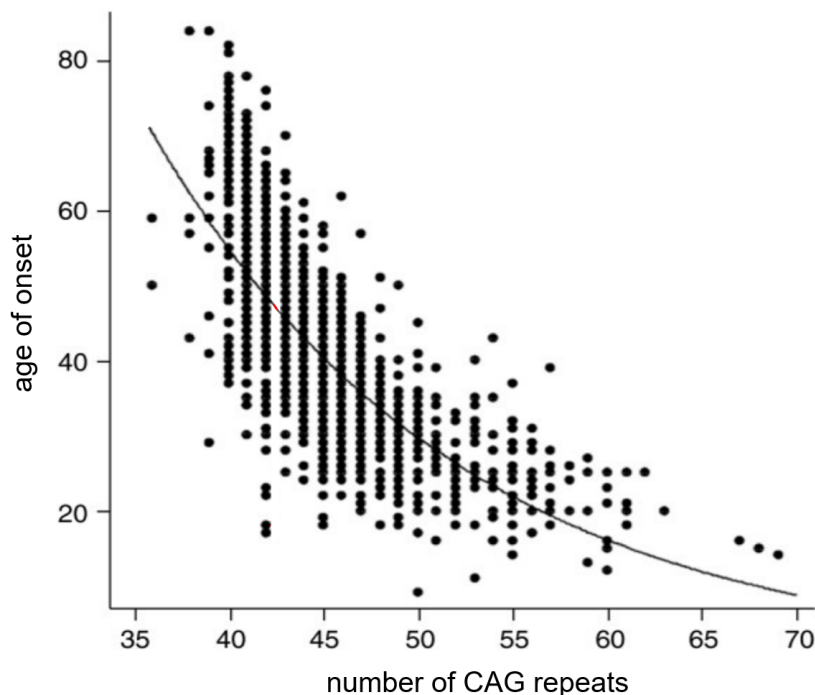


Fig. 2.2

- (i) Deduce if Huntington's disease is a dominant or recessive genetic disorder. [1]

- Dominant

- (ii) With reference to Fig. 2.2, evaluate the extent to which the number of CAG repeats can predict the age of onset of Huntington's disease. [3]

1. **[Cite general data]** Generally accurate, since as CAG repeats increases from 36 **[accept 35]** to 70, age at onset decreases from 70 to 11 **[accept 12]**.
2. **[Evaluate extent]** Not very accurate, since two persons with widely different age at onset can have the same number of repeats
3. **[Cite data]** e.g. 42 repeats, 18 and 75 years old / **other valid data**
4. **[Evaluate extent]** If no. of CAG repeats is 35 **[accept 36]** or less, definitely no Huntington disease, hence accurate.

[Any 3]

(iii) Explain how molecular techniques can be used to estimate the number of CAG repeats in total DNA isolated from a person with Huntington's disease. [4]

1. Design two different primers [**accept:** forward and reverse primers] flanking the CAG repeats
2. Perform polymerase chain reaction to amplify the CAG repeats
3. More repeats result in a longer fragment amplified.
4. Gel electrophoresis: separates amplified DNA fragment(s) based on their length
5. Estimate length of amplified product using the DNA ladder (of known fragment lengths)
6. [**Compulsory**] Divide the length by 3 to give the number of CAG repeats

Note: Deduct 1m for wrong sequence of procedure

[Any 4]

[Total: 19]



Section B
Answer **ONE** question.

Write your answers on the lined paper provided at the end of this Question Paper.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in parts (a) and (b), as indicated in the question.

QUESTION 3

(a) Discuss the biological importance of water to living organisms. [15]

[Role in photosynthesis and respiration]

1. Undergoes photolysis during (light-dependent stage of) photosynthesis
2. Source of electrons to replace electrons lost in photosystem II
3. Source of protons contributing to the high proton concentration in thylakoid space
4. Source of atmospheric oxygen as final electron acceptor in aerobic respiration
5. Forms when oxygen accepts electron and protons at end of the electron transport chain

[Role in reactions]

6. Hydrolysis reaction to break covalent bond
 - a. So that translation can be terminated by hydrolysis of acyl bond between polypeptide and tRNA / so that glucose released can be used as respiratory substrate, etc.
7. Restores the two polar groups (e.g. –OH and –OH/–NH₂)
8. Condensation reaction to form covalent bond
 - a. So that polysaccharides such as starch / glycogen can be used as energy storage molecules, etc.
9. **Valid example:** glycosidic bond / peptide bond / phosphodiester bond / ester bond

[Role as a solvent]

10. Solvent for hydrophilic molecules by forming hydrogen bonds / **named molecules** e.g. glucose, insulin, glucagon hormones, ions)
11. Solvent to dissolve gases / **named gases** e.g. oxygen, carbon dioxide
12. hence serves as a transport medium

[Role in driving formation of phospholipid bilayer]

13. Aqueous environment drives the phospholipids to adopt a bilayer structure
14. To shield the hydrophobic hydrocarbon chains away from water
15. Phosphate head interacts with water via hydrogen bond

[Role in driving protein folding]

16. Hydrophobic R-groups project into the protein core, interacting via hydrophobic interaction
17. Hydrophilic R-groups projects outwards, interacting with water via hydrogen bond
18. Allows globular proteins to be soluble in water
 - a. as water drives protein folding into its tertiary / 3D structure



19. **Named example:** haemoglobin / **AVP** [max 1]

20. Allows fibrous proteins to precipitate, because it is insoluble in water, fulfilling structural support role

21. **Named example:** collagen / keratin / spindle fiber / microtubule / **AVP**

[AVP]

22. Role in maintaining turgidity of plant / bacterial cells through osmotic pressure

23. Role in cooling in human by removal of latent heat via evaporation

24. Reactant in Krebs cycle e.g. converting fumarate to malate

25. Medium for removal of waste products in body via production of urine

26. Lubricant around joints

27. **AVP**

[Any 15]

- (b) Tryptophan synthesis and lactose catabolism are regulated in a similar manner in bacterial cells.

Describe how tryptophan synthesis in bacterial cells is regulated **and** explain the advantages of such a regulation system in the context of lactose catabolism. [10]

[How tryptophan synthesis is regulated] – at least 1

[*Trp operon*]

1. Repressible *Trp operon* system
2. Five structural genes / *Trp A-E* genes under control of one promoter
3. These genes code for enzymes that synthesize tryptophan

[*TrpR* regulatory gene]

4. *TrpR* gene codes for an inactive repressor protein
5. Inactive repressor not complementary in shape to and hence cannot bind to operator sequence

[When tryptophan is present]

6. Tryptophan is a co-repressor
7. Binds to and activates the repressor protein
8. Activated repressor proteins binds to the operator sequence
9. Blocks the RNA polymerase from binding to promoter
10. *Trp A-E* genes not transcribed

[When tryptophan is absent]

11. Unbound repressor dissociates from operator
12. RNA polymerase binds to promoter and transcribes *Trp A-E* genes

[Advantages of operon in the context of lactose catabolism] – at least 1

13. Allows genes that are involved in lactose catabolism to be regulated together
14. Allow cells to respond quickly to changes in environmental lactose concentration
15. Ensure cells do not synthesize these enzymes unnecessarily when lactose is absent

[Any 10]



QUESTION 4

(a) Explain why cancer development is a multi-step process **and** discuss the factors that increase of chances of cancer. [10]

[Why cancer development is a multi-step process] – at least 1

1. Gain-of-function mutation in proto-oncogenes that become oncogene
2. Loss-of-function mutation in tumor suppressor genes
3. One such mutation does not cause cancer immediately
4. A single cell needs to accumulate mutations in both tumor suppressor genes and proto-oncogenes to become cancerous
5. **Idea that** Accumulation of such mutations takes time
6. **Ref to** angiogenesis – tumour directs the growth of blood vessels to itself
7. **Ref to** metastasis – spread of some cancer cells from the primary tumour via the blood circulation to other body parts

AVP: Re-activation of telomerase gene needs to occur in order for cancer cells to divide indefinitely.

[Factors that increases of chances of cancer] – at least 1

8. Exposure to chemical carcinogens / **named carcinogen** e.g. ethidium bromide
9. Exposure to excessive ionizing radiation / **named ionizing radiation** e.g. X-rays
10. Exposure to ultraviolet radiation
11. Agents that cause inflammation, which generates DNA-damaging oxidizing agents in the cell
12. **Points 8-11:** cause DNA damage, resulting in replication error thus mutation
13. Infection by certain viruses / **named virus** e.g. human papillomavirus, HIV
14. Virus can introduce an oncogene
15. Virus can disrupt a tumor-suppressor gene during integration as provirus
16. Infection by certain bacteria / **named bacteria** e.g. *Helicobacter pylori*
17. **Ref to** Age
18. **Ref to** Genetic predisposition: inheriting an oncogene / mutated TSG from parent
19. **Ref to** Loss in immunity: cytotoxic T cells unable to destroy cancer cells

[Any 10]



- (b) The SARS-CoV-2 is an enveloped RNA virus that infects the respiratory airways, causing Covid-19 disease. Some Covid-19 vaccines contain mRNA that codes for a glycoprotein on the viral envelope.

Explain how intramuscular injection with the mRNA vaccine leads to the protection against SARS-CoV-2 **and** discuss the factors that affect the extent of transmission of the disease. [15]

[How vaccination with the mRNA vaccine leads to the protection] – max 9

uptake of mRNA and translation

1. mRNA taken into muscle / body cells
2. ribosomes translate the mRNA into the glycoprotein

presentation of antigen on surface of muscle cells via class I MHC to activate cytotoxic T cell

3. glycoprotein degraded into antigenic peptides and presented on class I MHC on the cell surface membrane of muscle cells
4. cytotoxic T cell with receptor complementary to the MHCI-peptide binds to the muscle cells and is activated (by cytokines secreted by helper T cells)
5. cytotoxic T cell secretes perforin and granzymes to kill muscle cells presenting MHCI-peptide

presentation of antigen on surface of APC via class II MHC activate helper T cell

6. glycoprotein secreted out of body/muscle cells
7. antigen presenting cells engulf the glycoprotein and digest it into smaller peptides
8. peptides are loaded onto class II MHC and presented on the cell surface membrane of APC
9. helper T cell with receptor complementary to the MHCII-peptide binds to the APC
10. APC secretes cytokines to activate the helper T cell, which then proliferates

activation of helper B cell

11. B cells with receptor specific for (the epitopes on) the glycoprotein binds, engulfs and present the peptide on MHCII
12. activated helper T cell binds to and activate B cells presenting the same peptide on MHCII
13. the B cells differentiate into plasma cells and memory B cells
14. plasma cells secrete antibodies against the glycoprotein

protection against SARS-CoV-2

15. memory B cells remain quiescent but will differentiate into plasma cells upon real infection with the virus
16. the antibodies secreted by plasma cells bind to the glycoprotein to prevent attachment of virus to the host cell surface receptor, preventing infection.



[Factors affecting spread] – max 6**17. Ref to evolution of the virus [1m for each sub-point]**

- a. How fast the virus evolves into different variants/strains...
- b. ...that can be more infectious / binds with higher affinity to host surface receptor
- c. ...that can result in a higher viral load / produce more viral progeny
- d. ...that can render the current vaccine ineffective

18. Ref to vaccine [1m for each sub-point]

- a. How fast the vaccine can be produced
- b. How fast the population is vaccinated to achieve individual protection
- c. How fast the population is vaccinated to achieve herd immunity
- d. Efficacy of the vaccine / % of vaccinated individuals who developed immunity

19. Ref to social responsibility [1m for each sub-point]

- a. Wearing of face masks
- b. Frequent washing of hands / use of hand sanitizers
- c. Keeping a safe distance
- d. Visiting doctor when having respiratory symptoms

20. Ref to population density – denser population leads to faster spread**21. Ref to actively testing people with respiratory symptoms****22. Ref to quarantining infected people****23. Ref to closing of country borders / restricting international travels****24. Ref to restricting movement/activity within a country****25. AVP****[Any 15]**😊 **END OF PAPER 3** 😊